0.2 Size of the Product of Subgroups

> **Theorem 0.2.1.** For $H, K \leq G$, if both are finite, then $|HK| = \dfrac{|H||K|}{|H \cap K|}$.

Proof. $HK = \bigcup_{h \in H} hK$. For $h_1, h_2 \in H$:

$$h_1K = h_2K \iff h_2^{-1}h_1 \in K \iff h_2^{-1}h_1 \in H \cap K \iff h_1(H \cap K) = h_2(H \cap K).$$

The number of distinct cosets hK is $|H : H \cap K|$. Since each coset has $|K|$ elements:

$$|HK| = |H : H \cap K| \cdot |K| = \frac{|H|}{|H \cap K|} \cdot |K|.$$

□

> **Theorem 0.2.2.** Let $H, K \leq G$ and $x \in HK$. Let $S = H \times K$. The size of the fiber
>
> $$|\{(h,k) \in H \times K \mid hk = x\}| = |H \cap K|.$$

Proof. Let $S = H \times K$. Define a relation $\sim$ on S as: $(h_1, k_1) \sim (h_2, k_2) \iff h_1k_1 = h_2k_2$.
This is an equivalence relation. Consider a partition $S/_\sim = \{[(h,k)] \mid h \in H, k \in K\}$.
For any class $[(h,k)]$, there exists a bijection $\varphi : [(1,1)] \to [(h,k)]$ given by:

$$\varphi(a, a^{-1}) = (ha, a^{-1}k).$$

Note that $(a, a^{-1}) \in [(1,1)]$ implies $a \in H \cap K$.
Injective: $(ha_1, a_1^{-1}k) = (ha_2, a_2^{-1}k) \implies a_1 = a_2$.
Surjective: Let $(h_1, k_1) \in S$ such that $h_1k_1 = hk$. Then $h^{-1}h_1 = kk_1^{-1} \in H \cap K$. Let this element be $a = h_1k_1$. Then,

$$(h_1, k_1) = (ha, a^{-1}k) = \varphi(a, a^{-1})$$

Consequently, since $|S| = |H||K|$ and each class has size $|H \cap K|$:

$$|S| = |H \cap K| \cdot (\text{number of classes}) = |H \cap K| \cdot |HK|.$$

□

0.3 Recognition Theorem for Direct Products

> **Theorem 0.3.1.** Suppose $H, K \trianglelefteq G$ are normal subgroups such that $H \cap K = \{1\}$ and $HK = G$. Then $G \cong H \times K$.

Proof. Define $\varphi : H \times K \to G$ by $\varphi(h,k) = hk$. We show φ is an isomorphism:

1. **Commutativity of elements:** Let $h \in H, k \in K$. Since $H, K \trianglelefteq G$,

   $$[h,k] = h^{-1}k^{-1}hk = (h^{-1}k^{-1}h)k \in K \quad \text{and} \quad [h,k] = h^{-1}(k^{-1}hk) \in H.$$

   Thus $[h,k] \in H \cap K = \{1\}$, which implies $hk = kh$.

2. **Homomorphism:**

   $$\begin{aligned} \varphi((h_1,k_1)(h_2,k_2)) &= \varphi(h_1h_2, k_1k_2) = h_1h_2k_1k_2 \\ &= h_1k_1h_2k_2 \quad (\text{since } h_2k_1 = k_1h_2) \\ &= \varphi(h_1,k_1)\varphi(h_2,k_2). \end{aligned}$$

3. **Injective:** $\varphi(h,k) = 1 \implies hk = 1 \implies h = k^{-1} \in H \cap K = \{1\}$. Thus $h = 1$ and $k = 1$.

4. **Surjective:** Follows from the assumption $HK = G$.

□